

LumenArc: Business and Technical Trade-off Analysis

PE6201 End-of-Course Project

Author: Ding Xiangfeng

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1 Problem and user

A digital asset research analyst often needs a precise number, definition, or market observation from a long report while market conditions and news continue to change. Searching manually is slow, but an invented answer is worse than a refusal because it can be mistaken for a trading signal. LumenArc is a research prototype for this analyst. Its core task is document-grounded question answering. It retrieves evidence from a named report, gives a source-linked answer, or states that the source does not provide enough evidence. It is not a trading system, a price forecaster, or personalised investment advice. The project originally committed to exactly thirty research PDFs. The final corpus contains thirty public monthly reports from one publisher. The reports were parsed into 559 pages and 1,017 searchable passages. Their source URLs, access records, checksums, and page counts are recorded in a manifest. The system can also parse user-supplied PDF, DOCX, text, and Markdown files. Original third-party PDF bytes are not redistributed because public access does not automatically grant redistribution rights.

2 Business and technical trade-offs

A keyword baseline was necessary because the problem is not solved merely by adding an AI model. The baseline returns the closest lexical passage. It is cheap, transparent, and useful when the requested wording appears exactly in the report. However, it cannot decide whether a nearby number answers the question, reconcile wording, or reliably refuse when no answer exists. It can return a plausible but irrelevant sentence. The final system therefore uses retrieval augmented generation rather than an unrestricted chat model. Retrieval constrains the model to selected passages, while citation and number checks constrain what may be labelled Supported.

The project owns the application-specific work: document extraction, page-aware chunking, lexical retrieval, numeric evidence verification, abstention rules, and evaluation. It rents the general language capability and external data feeds. This build versus rent decision avoids training a model on a small corpus and keeps the project focused on an analyst workflow. The cost is dependence on a model provider, request latency, uncertain usage cost, and the need to prevent provider output from being treated as truth. In the final evaluation, the protected school OpenRouter key was not confirmed, so the measured provider was Project AI rather than the school key. Spending is reported as unavailable, not zero or unlimited. Streamlit was considered because it enables a quick prototype. React with TypeScript and Node with Express were selected instead. This increased code volume, deployment work, and testing effort. In return, it gave a clearer frontend and backend boundary for the API key, explicit per-browser document-workspace state, real-time connection status, and a fuller bilingual interface. The server, rather than the browser, performs parsing, retrieval, external calls, and model requests. This is a deliberate security and product-state trade-off, not an unexplained framework change.

3 Implementation and data

The backend extracts text while retaining PDF page boundaries. It breaks pages into passages, ranks candidate passages for a question, and sends only those excerpts to the model. The model must return a structured answer and passage identifiers. A deterministic verifier checks that a Supported numeric answer has citations and that newly asserted numbers occur in cited evidence. The interface displays evidence states of Supported, Insufficient evidence, Conflicting evidence, or Needs clarification. It does not display a model-generated confidence percentage because no probability calibration was performed.

Live market prices and news are supporting features, not the main contribution. Ten Binance Spot pairs are shown as USDT quotes with connection and stale-data states. A one-time AI market pulse describes the displayed observations only and is not a forecast. News search uses current RSS and Google News results. Any optional AI news context is limited to the returned headline and excerpt. For AI document answers, selected excerpts leave the temporary server workspace for the external provider. Uploaded files are held in temporary memory without authentication or durable storage, so confidential documents are outside the prototype’s appropriate use.

4 Evaluation design

Twelve development questions were used to choose retrieval and verification behaviour. The final test used a frozen fifty-case holdout from different report months. It contains forty answerable numeric questions and ten report-specific no-answer cases. The holdout labels and configuration were checksummed before the formal run. The keyword baseline and final AI system received the same thirty parsed reports and the same questions.

A source-backed numeric answer counts only when the answer has the expected normalised number and a cited passage in the correct original report contains that number. A different valid page in the same report is accepted. This is a stricter test than number matching alone, but it is not full semantic validation. Correct refusal rate measures whether the system refuses the ten no-answer cases. Wrongful refusal rate measures whether it refuses an answerable case. Reporting both prevents an always-refusing system from looking safe and successful.

5 Results and critical analysis

Metric	Keyword baseline	Final AI system
Source-backed numeric answers out of 40	28 or 70.0 percent	37 or 92.5 percent
Correct refusals out of 10 no-answer cases	0 or 0 percent	10 or 100 percent
Wrongful refusals out of 40 answerable cases	2 or 5.0 percent	3 or 7.5 percent
Unsupported answers out of 10 no-answer cases	10 or 100 percent	0 or 0 percent

The final AI system improved source-backed numeric accuracy by 22.5 percentage points over the keyword baseline. It correctly refused all ten no-answer cases, while the baseline returned unsupported passages for all ten. This supports the value of retrieval plus a grounded-answer rule rather than lexical similarity alone. However, the outcome is not a claim that the system is generally correct. The evaluation checks numbers and cited report evidence. It does not prove that every explanation is complete, that every citation is the best citation, or that the approach transfers to other publishers.

The three wrongful refusals show the cost of conservative abstention. Two were retrieval misses within the correct monthly report. One case was labelled Conflicting evidence because the retrieved text contained both 20.2 percent and 13 percent declines for the queried period. Flagging a discrepancy can be responsible, but it still fails the answerable-case metric. The baseline had two wrongful refusals. Thus, the final system is safer on no-answer cases but not uniformly better at answering every supported case. Thirty-four final answers cited the label’s exact PDF page, while thirty-seven cited the correct report with the expected number. This difference shows why a citation metric should not be reduced to a single accuracy figure.

The final run took a median 3.55 seconds per case and a 90th-percentile 4.99 seconds per case. Two earlier incomplete runs stopped on malformed model JSON and produced no final summary. A format-only retry was added without changing the questions, retrieval method, model, answer rule, or evaluation scoring. This record matters because silently repeating a failed run after changing the system would weaken the credibility of the frozen holdout claim.

6 Limitations and conclusion

The labels were authored from original passages and machine-checked against source-page quotations, but they were not independently human-verified one by one. All thirty reports come from one publisher. PDF tables can lose layout during extraction, nearby numbers can be semantically irrelevant, and phrase search cannot prove that a no-answer fact is absent in every paraphrased form. No human analyst time study was conducted, so the project does not claim a 40 percent time saving.

LumenArc is therefore best understood as an evidence-first research prototype rather than a finished analyst product. Its strongest result is not the dashboard. It is the measured improvement in source-backed numeric answers, alongside explicit citations and tested abstention. Before production use, the next priorities would be independent label adjudication, a broader multi-publisher corpus, authenticated encrypted storage, cost controls, and a formal review of financial-harm scenarios.

Project Links

Final Project Video

<https://youtu.be/p8gAnGNghN0?si=KWcy5wya8AUTCqYM>

GitHub Repository

<https://github.com/Ding-hub6029/lumenarc-crypto-intelligence>